Experiment 1(f): 8-Bit Division Using 8085 Microprocessor

AIM:

To write an assembly language program to divide two 8-bit numbers stored in memory locations 4200H and 4201H, and store the remainder in 4202H and quotient in 4203H.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

ALGORITHM:

1. Load the divisor into register B.
2. Load the dividend into the accumulator.
3. Initialize register C to zero to count the quotient.
4. Compare the accumulator with the divisor.
5. If the accumulator is smaller than the divisor, jump to STOP.
6. Subtract the divisor from the accumulator.
7. Increment register C.
8. Repeat the process until the accumulator is smaller than the divisor.
9. Store the remainder at 4202H.
10. Store the quotient at 4203H.
11. Stop the program.

PROGRAM:

```
LDA 4201H
MOV B,A
LDA 4200H
MVI C,00H

LOOP: CMP B
JC STOP
SUB B
INR C
JMP LOOP
```

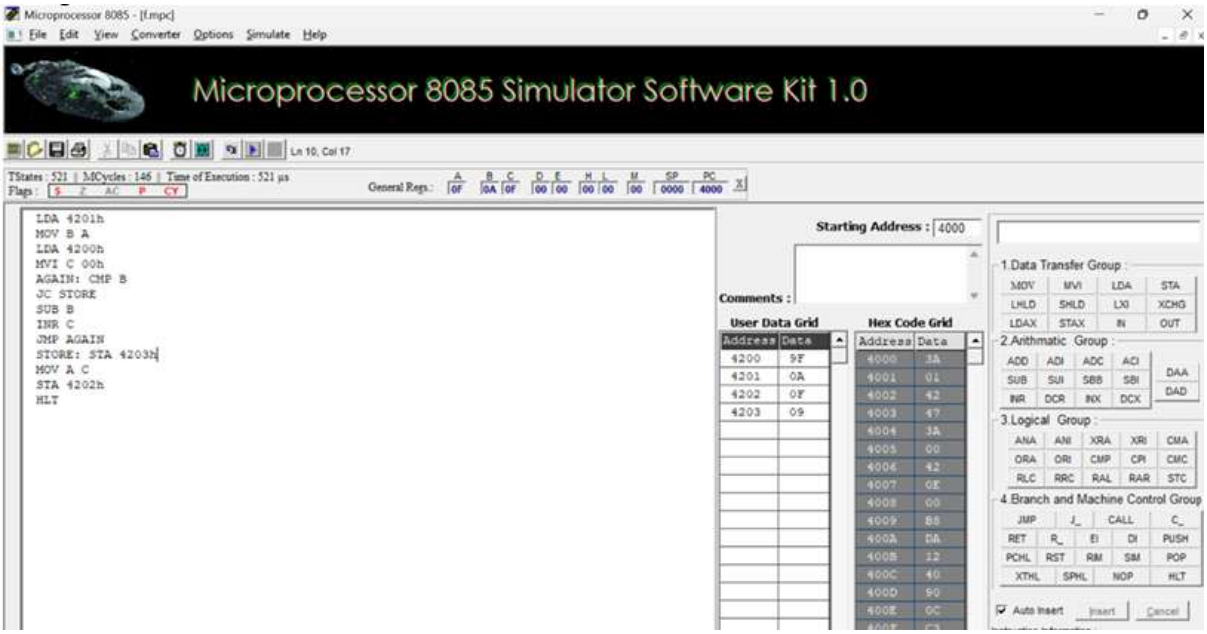
STOP: STA 4202H

MOV A,C

STA 4203H

HLT

OUTPUT:



RESULT:

Thus, an assembly language program to divide two 8-bit numbers using the 8085 microprocessor was written and executed successfully.